Problem Set 2

1

The power grid distributes current from a power station to the home using a series of long wires and transformer substations. A common misconception is that the term “Transmission Line” which is used to describe high voltage power lines, is the same as an electromagnetic transmission line. Consider these cases:

a

The new model of the toilet includes a small AI edge computing system to control a super-powered smart flusher. Unfortunately, this now means the toilet takes so much power it requires not only a plumber but an electrician to install. The Mains frequency in USA is 60 Hz. Assuming the neighborhood wires go from a local transformer to a home in a run of 500 m, can the wires be considered electrically short or long?

✓ **Answer**

$\lambda = \frac{v}{f} = \frac{3\times10^8}{60} = 5000\ \mathrm{km}$

$\frac{500\ \mathrm{m}}{5000\ \mathrm{km}} = 10^{-4}$

Since $0.0001 \ll 0.1$ this wire is considered very short.

b

Assuming the 500 m cable is a two-wire line, and the conductors are 2 mm in diameter, spaced at 4 mm, how much capacitance will this wire add to the electrical grid? Refer to Table 2-1 in the book and material properties for copper conductors.

✓ **Answer**

$C' = \frac{\pi\epsilon}{\ln\left(\left(\frac{D}{d}\right)+\sqrt{\left(\frac{D}{d}\right)^2-1}\right)}$

$D = 4\ \mathrm{mm}$

$d = 2\ \mathrm{mm}$

$L = 500\ \mathrm{m}$

$C' = 10.5606\ \mathrm{nF}$

c

One very rich client decides to run a dedicated line from the power plant to their smart toilet for minimal latency, and to avoid sharing with others. How far away can the power plant be before electromagnetic transmission line theory should be considered?

✓ **Answer**

$\lambda = 5000\ \mathrm{km}$

EM theory becomes relevant at around $\frac{\lambda}{10}$ or $\frac{\lambda}{20}$, which is as soon as $250 - 500\ \mathrm{km}$.

2 (2.2a)

A two-wire copper transmission line is embedded in a dielectric material with $\epsilon_r = 2.6$ and $\sigma = 2\times10^{-6}\ \mathrm{S/m}$. Its wires are separated by 3 cm and their radii are 1 mm each.

Calculate the line parameters R', L', G', and C' at $2\ \mathrm{GHz}$

✓ **Answer**

$\sigma_c = 5.8\times10^7\ \mathrm{S/m}$

$\mu_c = 0.99999\cdot(4\pi\times10^{-7})$

$R' = \frac{2\sqrt{\pi f\mu_c/\sigma_c}}{\pi d}$

$L' = \frac{\mu}{\pi}\ln\left(\left(\frac{D}{d}\right)+\sqrt{\left(\frac{D}{d}\right)^2-1}\right)$

$G' = \frac{\pi\sigma}{\ln\left(\left(\frac{D}{d}\right)+\sqrt{\left(\frac{D}{d}\right)^2-1}\right)}$

$C' = \frac{\pi\epsilon}{\ln\left(\left(\frac{D}{d}\right)+\sqrt{\left(\frac{D}{d}\right)^2-1}\right)}$

$R' = 3.714\ \Omega/\mathrm{m}$

$L' = 1.36\ \mu\mathrm{H/m}$

$G' = 1.848\ \mu\mathrm{S/m}$

$C' = 21.27\ \mathrm{pF/m}$

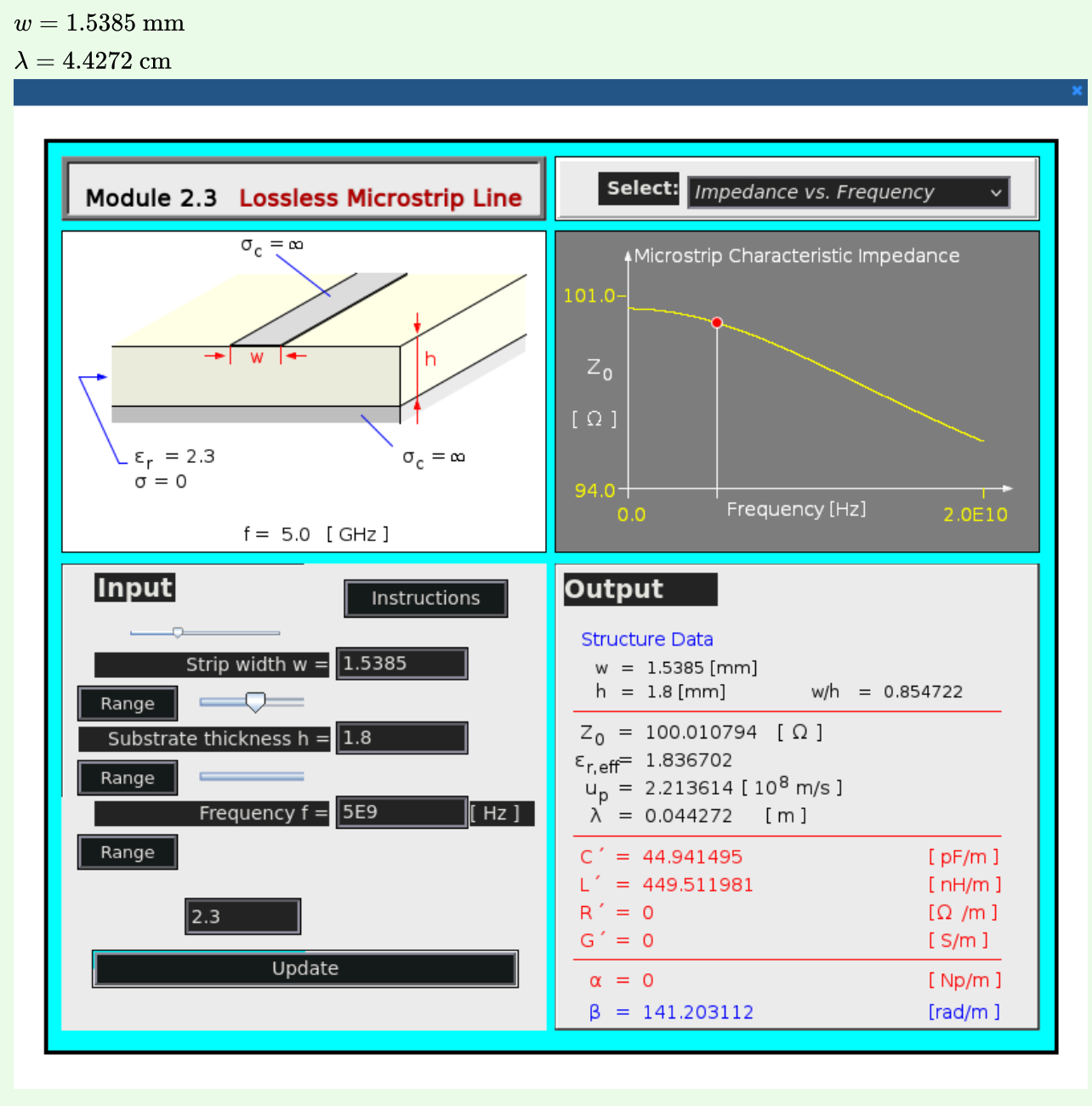
3 (2.10)

Use Module 2.3 to design a 100Ω microstrip transmission line. The substrate thickness is 1.8 mm and its $\epsilon_r = 2.3$. Select the strip width w, and determine the guide wavelength λ at $f = 5\ \mathrm{GHz}$. Include a printout of the screen display.

✓ **Answer**

$w = 1.5385\ \mathrm{mm}$

$\lambda = 4.4272\ \mathrm{cm}$



Strip width w = 1.5385

Substrate thickness h = 1.8

Frequency f = 5E9 [Hz]

2.3

Structure Data

w = 1.5385 [mm]

h = 1.8 [mm] w/h = 0.854722

Z0 = 100.010794 [Ω]

εr,eff = 1.836702

up = 2.213614 [10⁸ m/s]

λ = 0.044272 [m]

C´ = 44.941495 [pF/m]

L´ = 449.511981 [nH/m]

R´ = 0 [Ω /m]

G´ = 0 [S/m]

α = 0 [Np/m]

β = 141.203112 [rad/m]

4 (2.17)

Using a slotted line, the voltage on a lossless transmission line was found to have a maximum magnitude of 1.5 V and a minimum magnitude of 0.6 V. Find the magnitude of the load’s reflection coefficient.

✓ **Answer**

$S = \frac{V_{max}}{V_{min}} = 2.5$

$\Gamma = \frac{S-1}{S+1} = 0.4286$

5 (2.19)

A 50Ω lossless transmission line is terminated in a load with impedance $Z_L = (30 - j50)\Omega$. The wavelength is 8 cm.

Determine:

a

The reflection coefficient at the load.

✓ **Answer**

$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

$\Gamma = 0.10112359550561797 - 0.5617977528089888j$

$\Gamma = 0.5708263279047433\angle - 79.79602627826831^\circ$

b

The standing-wave ratio on the line.

✓ **Answer**

$S = \frac{1+|\Gamma|}{1-|\Gamma|}$

$S = 3.6601181061174053$

c

The position of the voltage maximum nearest the load.

✓ **Answer**

$d_{max} = \frac{\theta_r\lambda}{4\pi} + \frac{n\lambda}{2}$

$\theta_r = -1.392703388563699$

$d_{max} = 3.11\ \mathrm{cm}$

d

The position of the current maximum nearest the load.

✓ **Answer**

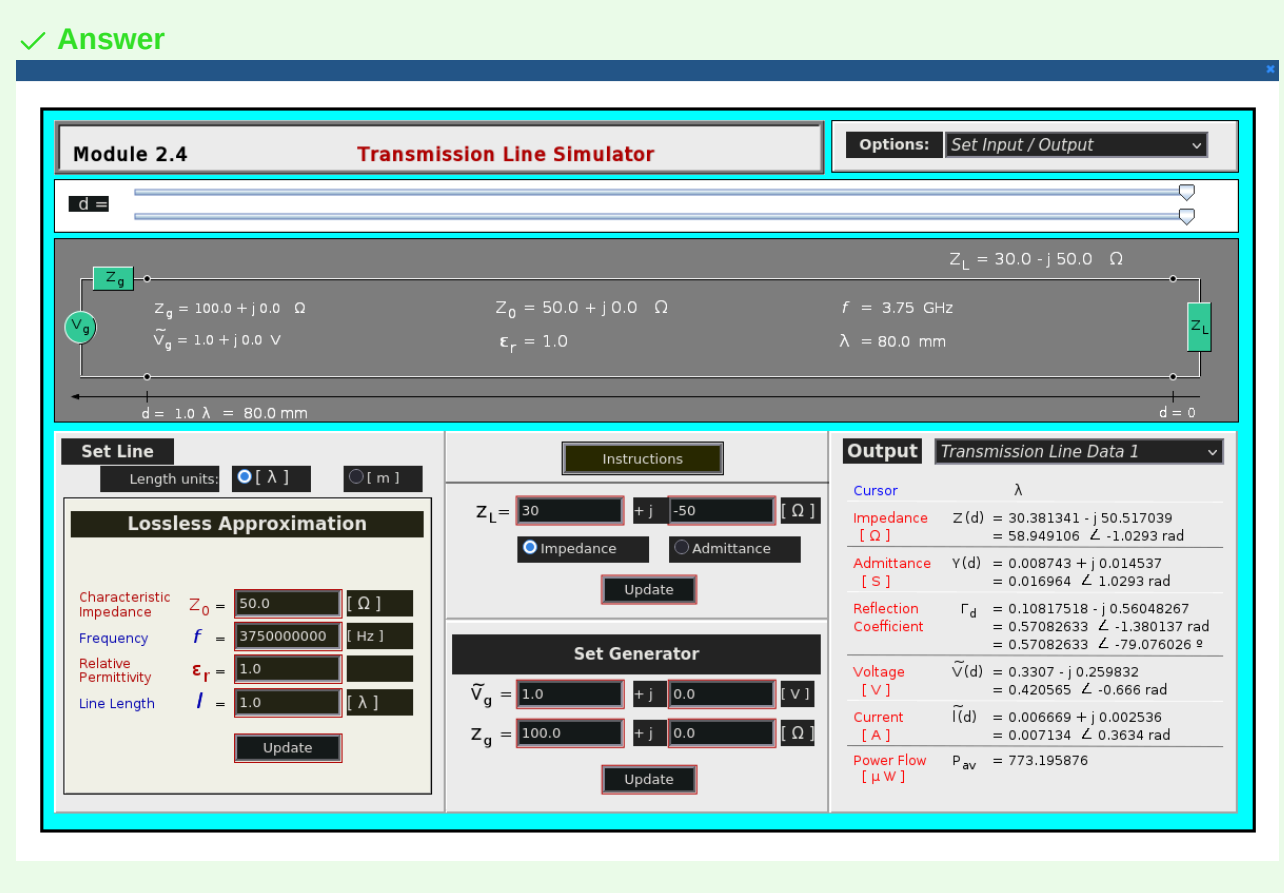
$I_{max} = d_{max} - \frac{\lambda}{4}$

$I_{max} = 1.11\ \mathrm{cm}$

e

Verify quantities in parts (a)–(d) using Module 2.4. Include a printout of the screen display.

✓ **Answer**



ZL = 30.0 - j 50.0 Ω

Zg = 100.0 + j 0.0 Ω

Vg = 1.0 + j 0.0 V

Z0 = 50.0 + j 0.0 Ω

εr = 1.0

f = 3.75 GHz

λ = 80.0 mm

d = 1.0 λ = 80.0 mm

d = 0

Length units: ● [λ] ○ [m]

Characteristic Impedance Z0 = 50.0 [Ω]

Frequency f = 3750000000 [Hz]

Relative Permittivity εr = 1.0

Line Length l = 1.0 [λ]

ZL = 30 + j -50 [Ω]

● Impedance ○ Admittance

Vg = 1.0 + j 0.0 [V]

Zg = 100.0 + j 0.0 [Ω]

Cursor λ

Impedance [Ω] Z(d) = 30.381341 - j 50.517039 = 58.949106 ∠ -1.0293 rad

Admittance [S] Y(d) = 0.008743 + j 0.014537 = 0.016964 ∠ 1.0293 rad

Reflection Coefficient Γd = 0.10817518 - j 0.56048267 = 0.57082633 ∠ -1.380137 rad = 0.57082633 ∠ -79.076026 º

Voltage [V] V(d) = 0.3307 - j 0.259832 = 0.420565 ∠ -0.666 rad

Current [A] I(d) = 0.006669 + j 0.002536 = 0.007134 ∠ 0.3634 rad

Power Flow [μ W] Pav = 773.195876